MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

INTRODUCTION TO TRIGONOMETRY

Ratios, Exact Values, Transformations & Pythagorean Identities

Academic Year 2026–27

Master the Acute Angle. Understand the Invariance.
Calculate with Absolute Precision.

6 Trigonometric Ratios | k-Multiplier Protocol | Special Angles | Pythagorean Identities

A publication-quality Maths Manthan learning and revision resource



What this handbook includes

- Complete geometric foundations: Opposite (Perpendicular), Adjacent (Base), Hypotenuse, and Orientation Shift.
- Scale Invariance Theorem: Why trigonometric ratios depend solely on angle θ (Thales/AA similarity proof).
- The k-multiplier evaluation method for finding all ratios from one.
- Geometric derivations of $30°, 45°, 60°$ and boundaries $0°, 90°$.
- Complete Pythagorean identity derivations and master proving toolkits.

How to use it for maximum marks

Always sketch a neat right-angled triangle and locate the acute reference angle first. When given a numerical ratio (e.g. $\sin A = 3/5$), declare the side lengths with a positive proportionality multiplier k $(3k, 5k)$ to protect your method marks. In identity proofs, quote the standard Pythagorean formulas in parentheses alongside each transformation step.

Official syllabus scope (CBSE Class X 2026–27)

The published CBSE Class X curriculum for 2026–27 prescribes **Trigonometric ratios of an acute angle of a right-angled triangle. Proof of their existence (well defined); values of the trigonometric ratios of** $30°, 45°, 60°$**. Relationships between the ratios. Proof and applications of the identity** $\sin^2 A + \cos^2 A = 1$**. Only simple identities to be given.**

Strictly Excluded Topics (Deleted from Rationalized Syllabus):
Section 8.4: Trigonometric Ratios of Complementary Angles ($\sin(90° - A) = \cos A$, $\tan(90° - A) = \cot A$, etc.) and Exercise 8.3 of older textbooks are **completely deleted**. Questions directly requiring complementary angle identities are excluded from the board examination.

Source: CBSE Mathematics Class X Curriculum (2026–27), Unit V: Trigonometry, Chapter 8.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 8

CORE SYLLABUS

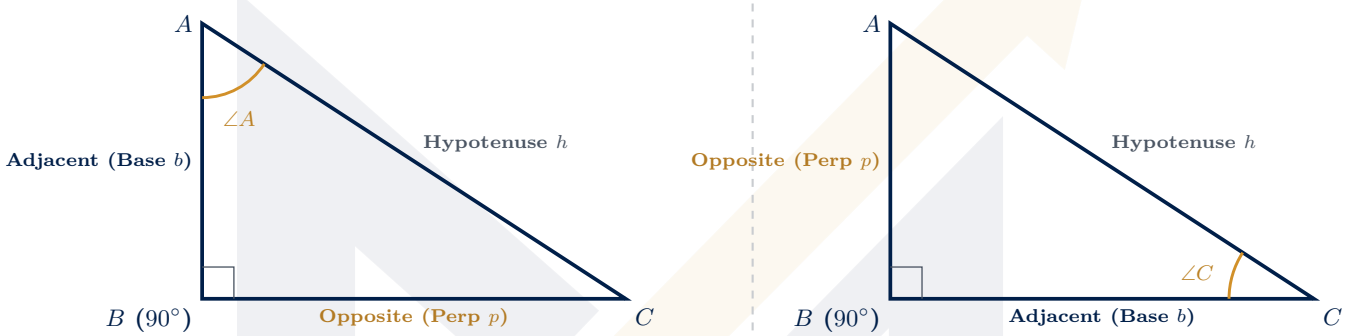
GEOMETRIC FOUNDATIONS

Right Triangle Anatomy, Reference Angle Shift & Scale Invariance

Module 1 | Anatomy of a Right Triangle & The Reference Angle Shift

Trigonometry begins with the right-angled triangle, where side names are strictly relative to the acute angle under consideration:

- **Hypotenuse (h):** The side opposite to the right angle ($90°$). It is strictly the longest side of the triangle.
- **Opposite Side / Perpendicular (p):** The side directly opposite to the selected acute reference angle θ.
- **Adjacent Side / Base (b):** The side that forms the acute angle θ alongside the hypotenuse.



Module 2 | The Scale Invariance Theorem: Geometric Well-Definedness

Proof of Existence (Well-Definedness of Ratios via AA Similarity)

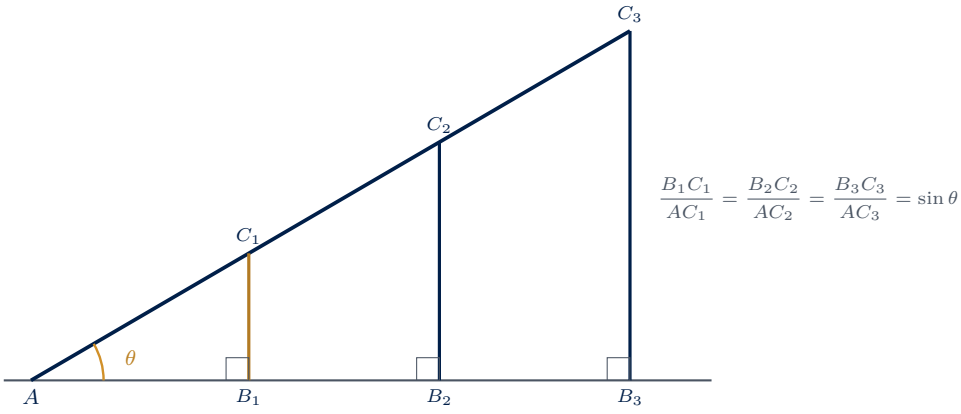
Fundamental Theorem: The trigonometric ratios of an acute angle θ depend **only** on the measure of angle θ, and are entirely independent of the side lengths or physical dimensions of the triangle.
Geometric Proof: Consider two right-angled triangles $\triangle AB_1C_1$ and $\triangle AB_2C_2$ sharing acute angle $\angle A = \theta$:

$$\angle A = \angle A \quad \text{(Common)}, \qquad \angle B_1 = \angle B_2 = 90°$$

By the AA (Angle-Angle) Similarity Criterion: $\mathbf{\triangle AB_1C_1 \sim \triangle AB_2C_2}$.
Therefore, their corresponding sides are strictly proportional:

$$\frac{\mathbf{B_1C_1}}{\mathbf{B_2C_2}} = \frac{\mathbf{AC_1}}{\mathbf{AC_2}} = \frac{\mathbf{AB_1}}{\mathbf{AB_2}} \iff \frac{\mathbf{B_1C_1}}{\mathbf{AC_1}} = \frac{\mathbf{B_2C_2}}{\mathbf{AC_2}} = \sin\theta$$

Thus, whether measured in millimeters or kilometers, the ratio $\frac{\text{Opposite}}{\text{Hypotenuse}}$ is an invariant constant for a given angle θ.



CHAPTER 8

CORE SYLLABUS

THE SIX TRIGONOMETRIC RATIOS

Definitions, Reciprocal Pairs & Quotient Relations

Module 1 | Formal Geometric Definitions

For an acute angle θ in a right-angled triangle with Opposite (p), Adjacent (b), and Hypotenuse (h):

Ratio	Symbol	Geometric Ratio	Intuition & Physical Meaning
Sine	$\sin\theta$	$\frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{p}{h}$	Fractional vertical elevation achieved per unit hypotenuse.
Cosine	$\cos\theta$	$\frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{b}{h}$	Fractional horizontal base covered per unit hypotenuse.
Tangent	$\tan\theta$	$\frac{\text{Opposite}}{\text{Adjacent}} = \frac{p}{b}$	Geometric steepness or slope (rise over run) of the incline.
Cosecant	$\csc\theta$	$\frac{\text{Hypotenuse}}{\text{Opposite}} = \frac{h}{p}$	Reciprocal of sine: total travel needed per unit vertical gain.
Secant	$\sec\theta$	$\frac{\text{Hypotenuse}}{\text{Adjacent}} = \frac{h}{b}$	Reciprocal of cosine: total travel needed per unit horizontal span.
Cotangent	$\cot\theta$	$\frac{\text{Adjacent}}{\text{Opposite}} = \frac{b}{p}$	Reciprocal of tangent: horizontal run per unit vertical height.

Module 2 | Reciprocal Pairs & Quotient Relations

Governing Algebraic Inter-Relations

Reciprocal Identities (Product Equals 1):

$\sin\theta = \frac{1}{\csc\theta} \iff \sin\theta \cdot \csc\theta = \mathbf{1}$ | $\cos\theta = \frac{1}{\sec\theta} \iff \cos\theta \cdot \sec\theta = \mathbf{1}$ | $\tan\theta = \frac{1}{\cot\theta} \iff \tan\theta \cdot \cot\theta = \mathbf{1}$

Quotient Identities (Ratios of Primary Functions):

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\mathbf{p/h}}{\mathbf{b/h}} = \mathbf{\frac{p}{b}}$ $\qquad$ $\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{\mathbf{b/h}}{\mathbf{p/h}} = \mathbf{\frac{b}{p}}$

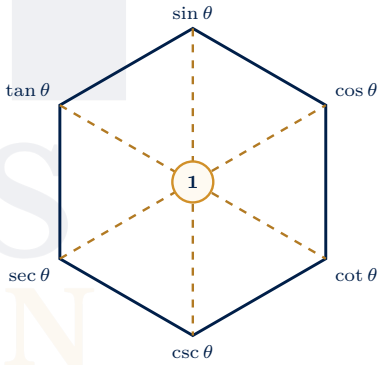
Module 3 | Acute Bounds ($0^\circ < \theta < 90^\circ$)

Strict Mathematical Bounds

Hypotenuse is always strictly the longest side ($h > p$ and $h > b$). Consequently for all acute θ:

- $\mathbf{0} < \sin\theta < \mathbf{1}$ and $\mathbf{0} < \cos\theta < \mathbf{1}$ (**Never** > 1).
- $\sec\theta > \mathbf{1}$ and $\csc\theta > \mathbf{1}$ (**Never** < 1).
- $\mathbf{0} < \tan\theta < \infty$ and $\mathbf{0} < \cot\theta < \infty$.

Trap Alert: $\sin\theta$ and $\cos\theta$ can **never** exceed 1!



Opposite Diagonals: $\times = 1$

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CHAPTER 8

WORKED MODEL

RATIO COMPUTATION PROTOCOLS

Given One Ratio, Finding All Others: The k-Factor Method

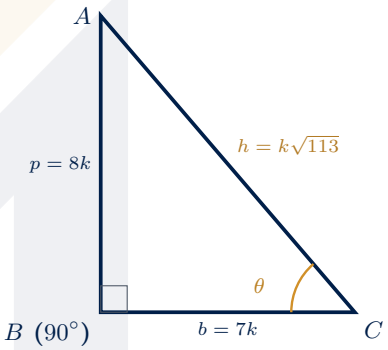
Module 1 | The Mandatory k-Proportionality Rule for Board Exams

Why CBSE Examiners Deduct Marks for Missing 'k'

Critical Concept: A trigonometric ratio represents the **relative scale** between two sides, not their absolute physical lengths. For instance, $\sin A = \frac{3}{5}$ means the ratio is $3:5$; sides could be $3\,\text{cm}, 5\,\text{cm}$ or $6\,\text{cm}, 10\,\text{cm}$.
Official Marking Scheme Rule: Writing Opposite $= 3$ and Hypotenuse $= 5$ without defining a positive real scaling multiplier $k > 0$ violates dimensional rigor and risks a **0.5 to 1 mark penalty**. Always state:

$$\text{Opposite} = \mathbf{3k} \quad \text{and} \quad \text{Hypotenuse} = \mathbf{5k}, \quad \text{where } \mathbf{k > 0}$$

Module 2 | Worked Board Archetype 1: Expression Evaluation with Given Cotangent



Problem: If $\cot\theta = \frac{7}{8}$, evaluate: (i) $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$, (ii) $\cot^2\theta$.
Solution:
Step 1: Declare side lengths with k.
In right $\triangle ABC$, $\cot\theta = \frac{b}{p} = \frac{7}{8}$. Let $b = 7k, p = 8k$ $(k > 0)$.
Step 2: Hypotenuse by Pythagoras Theorem.
$h = \sqrt{(8k)^2 + (7k)^2} = \sqrt{64k^2 + 49k^2} = k\sqrt{113}$.
Step 3: Algebraic Identity Simplification.
$\text{Num} = (1+\sin\theta)(1-\sin\theta) = 1 - \sin^2\theta = \cos^2\theta$
$\text{Den} = (1+\cos\theta)(1-\cos\theta) = 1 - \cos^2\theta = \sin^2\theta$
$\therefore \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \left(\frac{7}{8}\right)^2 = \mathbf{\frac{49}{64}}$. Both (i) and (ii) evaluate to $\mathbf{\frac{49}{64}}$.

Module 3 | Worked Board Archetype 2: Coefficient on Ratio ($15\cot A = 8$)

Worked Model: Isolate the Ratio First

Problem: Given $15\cot A = 8$, find the values of $\sin A$ and $\sec A$.
Solution:
Step 1: Isolate ratio. $\cot A = \frac{8}{15} = \frac{\text{Adjacent } b}{\text{Opposite } p}$. Let $b = 8k$ and $p = 15k$ $(k > 0)$.
Step 2: Hypotenuse. $h = \sqrt{(15k)^2 + (8k)^2} = \sqrt{225k^2 + 64k^2} = \sqrt{289k^2} = 17k$.
Step 3: State desired ratios.
$\sin\mathbf{A} = \frac{\text{Opposite}}{\text{Hypotenuse}} = \mathbf{\frac{15k}{17k}} = \mathbf{\frac{15}{17}}$ $\qquad$ $\sec\mathbf{A} = \frac{\text{Hypotenuse}}{\text{Adjacent}} = \mathbf{\frac{17k}{8k}} = \mathbf{\frac{17}{8}}$

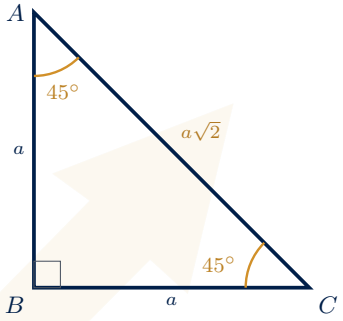
CHAPTER 8

CORE SYLLABUS

STANDARD ANGLE GEOMETRY

Geometric Derivations of 45°, 30°, and 60° Ratios

Module 1 | Geometric Derivation of Ratios of 45° via Isosceles Right Triangle



Construction: Consider an isosceles right $\triangle ABC$ right-angled at B with $\angle A = 45°$.

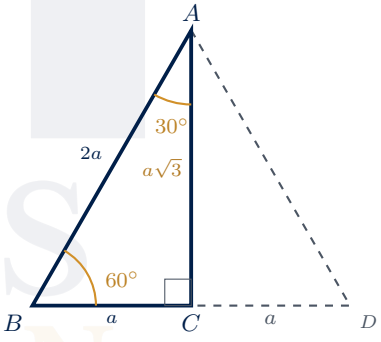
- By Angle Sum Property: $\angle C = 180° - (90° + 45°) = 45°$.
- Equal angles subtend equal sides: $AB = BC = a$.
- By Pythagoras Theorem:

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = a\sqrt{2}$$

$$\sin \mathbf{45°} = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}}, \qquad \cos \mathbf{45°} = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}}, \qquad \tan \mathbf{45°} = \mathbf{1}$$

(Reciprocals: $\csc 45° = \sqrt{2}$, $\sec 45° = \sqrt{2}$, $\cot 45° = 1$)

Module 2 | Geometric Derivation of Ratios of 30° and 60° via Equilateral Triangle



Construction: Let $\triangle ABD$ be an equilateral triangle with side $2a$. Each angle $= 60°$. Draw altitude $AC \perp BD$.

- Altitude bisects base: $BC = CD = a$.
- Altitude bisects vertical angle: $\angle BAC = 30°$.
- In right $\triangle ABC$, by Pythagoras Theorem:

$$AC = \sqrt{(2a)^2 - a^2} = \sqrt{4a^2 - a^2} = \sqrt{3a^2} = a\sqrt{3}$$

Ratios for $30°$ **(**$p = a$**,** $b = a\sqrt{3}$**,** $h = 2a$**):**

$$\sin \mathbf{30°} = \frac{\mathbf{1}}{\mathbf{2}}, \qquad \cos \mathbf{30°} = \frac{\sqrt{\mathbf{3}}}{\mathbf{2}}, \qquad \tan \mathbf{30°} = \frac{\mathbf{1}}{\sqrt{\mathbf{3}}}$$

Ratios for $60°$ **(**$p = a\sqrt{3}$**,** $b = a$**,** $h = 2a$**):**

$$\sin \mathbf{60°} = \frac{\sqrt{\mathbf{3}}}{\mathbf{2}}, \qquad \cos \mathbf{60°} = \frac{\mathbf{1}}{\mathbf{2}}, \qquad \tan \mathbf{60°} = \sqrt{\mathbf{3}}$$

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CHAPTER 8

CORE SYLLABUS

EXACT VALUES ENGINE

The Master Angle Table, Monotonicity & Mental Shortcuts

Module 1 | The Master Trigonometric Table for Standard Angles

Ratio / Angle	0°	30°	45°	60°	90°
$\sin\theta$	$\mathbf{0}$	$\mathbf{\frac{1}{2}}$	$\mathbf{\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}}$	$\mathbf{\frac{\sqrt{3}}{2}}$	$\mathbf{1}$
$\cos\theta$	$\mathbf{1}$	$\mathbf{\frac{\sqrt{3}}{2}}$	$\mathbf{\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}}$	$\mathbf{\frac{1}{2}}$	$\mathbf{0}$
$\tan\theta$	$\mathbf{0}$	$\mathbf{\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}}$	$\mathbf{1}$	$\mathbf{\sqrt{3}}$	**Not Defined** (∞)
$\csc\theta$	**Not Defined**	$\mathbf{2}$	$\mathbf{\sqrt{2}}$	$\mathbf{\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}}$	$\mathbf{1}$
$\sec\theta$	$\mathbf{1}$	$\mathbf{\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}}$	$\mathbf{\sqrt{2}}$	$\mathbf{2}$	**Not Defined** (∞)
$\cot\theta$	**Not Defined**	$\mathbf{\sqrt{3}}$	$\mathbf{1}$	$\mathbf{\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}}$	$\mathbf{0}$

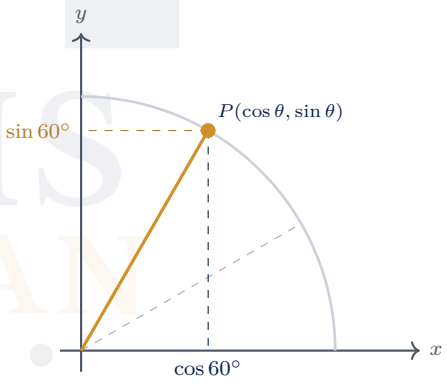
Module 2 | The "Square Root of Quarters" Mental Memory Trick

5-Second Table Reconstruction Protocol

Write the whole numbers $0, 1, 2, 3, 4$ under angles $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$, divide each by 4, and take the square root:

$$\sin 0^\circ = \sqrt{\frac{0}{4}} = \mathbf{0}, \quad \sin 30^\circ = \sqrt{\frac{1}{4}} = \mathbf{\frac{1}{2}}, \quad \sin 45^\circ = \sqrt{\frac{2}{4}} = \mathbf{\frac{1}{\sqrt{2}}}, \quad \sin 60^\circ = \sqrt{\frac{3}{4}} = \mathbf{\frac{\sqrt{3}}{2}}, \quad \sin 90^\circ = \sqrt{\frac{4}{4}} = \mathbf{1}$$

To find $\cos\theta$, simply reverse the row order: $1, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}, 0$. Then $\tan\theta = \frac{\sin\theta}{\cos\theta}$.



Unit Radius Invariant

Module 3 | Monotonicity Trajectories ($0^\circ \leq \theta \leq 90^\circ$)

Behavior as Angle θ Increases

- $\sin\theta$ **strictly increases** from 0 to 1.
- $\cos\theta$ **strictly decreases** from 1 to 0.
- $\tan\theta$ **strictly increases** from 0 to ∞.
- At $\theta = \mathbf{45}^\circ$: $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2}$.
- For $\theta < \mathbf{45}^\circ$: $\cos\theta > \sin\theta$.
- For $\theta > \mathbf{45}^\circ$: $\sin\theta > \cos\theta$.

CHAPTER 8

WORKED MODEL

NUMERICAL EVALUATION

Order of Operations, Quadratic Powers & Linear Systems

Module 1 | Power Notation & Evaluation Rules

Standard Power Notation Protocol

$$\sin^{\mathbf{2}}\theta \equiv (\sin\theta)^{\mathbf{2}}, \qquad \cos^{\mathbf{n}}\theta \equiv (\cos\theta)^{\mathbf{n}}, \qquad \tan^{\mathbf{2}}\theta \equiv (\tan\theta)^{\mathbf{2}}$$

Fatal Notation Trap: $\sin^2\theta$ does **NOT** mean $\sin(\theta^2)$. For example, $\sin^2 30^\circ = (\sin 30^\circ)^2 = (1/2)^2 = 1/4$, whereas $\sin(30^\circ)^2 = \sin(900^\circ)$ which is an entirely different quantity!

Module 2 | Worked Board Archetype 1: Multi-Term Surd Evaluation

Worked Model: Comprehensive Fractional Evaluation

Problem: Evaluate the exact numerical value of:

$$E = \frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$$

Solution:

Step 1: Substitute exact values into numerator.

$\cos 60^\circ = \frac{1}{2} \implies \cos^2 60^\circ = \frac{1}{4}$; $\sec 30^\circ = \frac{2}{\sqrt{3}} \implies \sec^2 30^\circ = \frac{4}{3}$; $\tan 45^\circ = 1 \implies \tan^2 45^\circ = 1$.

Numerator $= 5\left(\frac{1}{4}\right) + 4\left(\frac{4}{3}\right) - 1 = \frac{5}{4} + \frac{16}{3} - 1 = \frac{15+64-12}{12} = \frac{67}{12}$.

Step 2: Evaluate denominator (Pythagorean identity confirmation).

Denominator $= \sin^2 30^\circ + \cos^2 30^\circ = \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} + \frac{3}{4} = 1$.

Step 3: Compute final quotient.

$$E = \frac{67/12}{1} = \mathbf{\frac{67}{12}} = \mathbf{5\frac{7}{12}}$$

Module 3 | Worked Board Archetype 2: System of Trigonometric Equations

Worked Model: Determining Acute Angles A and B

Problem: If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find acute angles A and B.

Solution:

Step 1: Translate trigonometric equations into linear equations.

$\tan(A+B) = \sqrt{3} = \tan 60^\circ \implies \mathbf{A + B = 60^\circ}$ — (1); $\tan(A-B) = \frac{1}{\sqrt{3}} = \tan 30^\circ \implies \mathbf{A - B = 30^\circ}$ — (2).

Step 2: Solve simultaneously by elimination.

Adding (1) and (2): $2A = 90^\circ \implies \mathbf{A = 45^\circ}$. Subtracting (2) from (1): $2B = 30^\circ \implies \mathbf{B = 15^\circ}$.

Final Statement: $A = \mathbf{45^\circ}$ and $B = \mathbf{15^\circ}$. Both are acute and satisfy $A + B = 60^\circ \leq 90^\circ$.

CHAPTER 8

CORE SYLLABUS

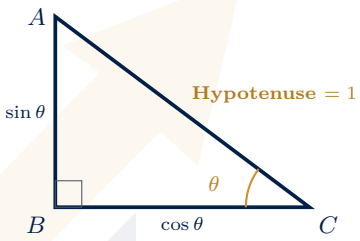
PYTHAGOREAN IDENTITIES

Geometric Derivation, Branching Formulas & Conjugates

Module 1 | Mathematical Meaning of an Identity

An equation is called an **identity** when it is true for all values of the variables for which the functions are defined.

Module 2 | Geometric Derivation of the Primary Pythagorean Identity



Consider right $\triangle ABC$ right-angled at B with acute angle $\angle A = \theta$:

- By Pythagoras Theorem:

$$AB^2 + BC^2 = AC^2$$

- Dividing each term by AC^2:

$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2} \implies \left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = 1$$

- Since $\frac{AB}{AC} = \cos A$ and $\frac{BC}{AC} = \sin A$:

$$\boxed{\sin^2 \mathbf{A} + \cos^2 \mathbf{A} = \mathbf{1}}$$

Module 3 | The Three Sibling Identities & Difference-of-Squares Branching

Master Identity Triad & Algebraic Conjugate Factors

1. Sine-Cosine Identity: Valid for all acute angles $0° \le \theta \le 90°$:

$$\sin^2\theta + \cos^2\theta = \mathbf{1} \iff \sin^2\theta = \mathbf{1} - \cos^2\theta \iff \cos^2\theta = \mathbf{1} - \sin^2\theta$$

2. Tangent-Secant Identity: Valid for $0° \le \theta < 90°$ (dividing primary identity by $\cos^2\theta$):

$$\mathbf{1} + \tan^2\theta = \sec^2\theta \iff \sec^2\theta - \tan^2\theta = \mathbf{1}$$

$$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = \mathbf{1} \iff \sec\theta - \tan\theta = \frac{\mathbf{1}}{\sec\theta + \tan\theta}$$

3. Cotangent-Cosecant Identity: Valid for $0° < \theta \le 90°$ (dividing primary identity by $\sin^2\theta$):

$$\mathbf{1} + \cot^2\theta = \csc^2\theta \iff \csc^2\theta - \cot^2\theta = \mathbf{1}$$

$$(\csc\theta - \cot\theta)(\csc\theta + \cot\theta) = \mathbf{1} \iff \csc\theta - \cot\theta = \frac{\mathbf{1}}{\csc\theta + \cot\theta}$$

Golden Exam Hack: If $\sec\theta + \tan\theta = x$, then instantly $\sec\theta - \tan\theta = \frac{1}{x}$, allowing immediate linear addition to solve for $\sec\theta = \frac{1}{2}(x + \frac{1}{x})$!

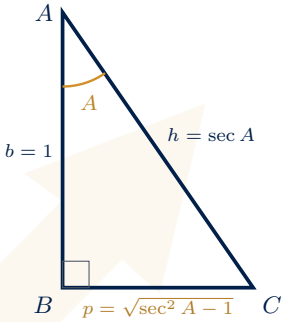
CHAPTER 8

REFERENCE MATRIX

RATIO CONVERSION MATRIX

Expressing Any Trigonometric Ratio in Terms of Any Other

Module 1 | The Parameterized Reference Triangle Method



Pedagogical Strategy: The fastest, safest board-exam technique to express all trigonometric ratios in terms of a single target ratio (e.g. $\sec A$) is constructing a **parameterized right triangle**.

To write all ratios in terms of sec **A**:

- Set $\sec A = \frac{\sec A}{1} = \frac{\text{Hypotenuse}}{\text{Adjacent (Base)}}$.
- Let Base $= 1$ and Hypotenuse $= \sec A$.
- By Pythagoras Theorem:

$$\text{Opposite } p = \sqrt{\sec^2 A - 1}$$

Every ratio is now read directly from the triangle!

Module 2 | Master 6×6 Universal Ratio Transformation Matrix

All ratios for an acute angle A expressed in terms of each column's designated generator:

Ratio	In terms of $\sin A$	In terms of $\cos A$	In terms of $\tan A$	In terms of $\sec A$
$\sin$ **A**	$\sin A$	$\sqrt{1-\cos^2 A}$	$\dfrac{\tan A}{\sqrt{1+\tan^2 A}}$	$\dfrac{\sqrt{\sec^2 A - 1}}{\sec A}$
$\cos$ **A**	$\sqrt{1-\sin^2 A}$	$\cos A$	$\dfrac{1}{\sqrt{1+\tan^2 A}}$	$\dfrac{1}{\sec A}$
$\tan$ **A**	$\dfrac{\sin A}{\sqrt{1-\sin^2 A}}$	$\dfrac{\sqrt{1-\cos^2 A}}{\cos A}$	$\tan A$	$\sqrt{\sec^2 A - 1}$
$\csc$ **A**	$\dfrac{1}{\sin A}$	$\dfrac{1}{\sqrt{1-\cos^2 A}}$	$\dfrac{\sqrt{1+\tan^2 A}}{\tan A}$	$\dfrac{\sec A}{\sqrt{\sec^2 A - 1}}$
$\sec$ **A**	$\dfrac{1}{\sqrt{1-\sin^2 A}}$	$\dfrac{1}{\cos A}$	$\sqrt{1+\tan^2 A}$	$\sec A$
$\cot$ **A**	$\dfrac{\sqrt{1-\sin^2 A}}{\sin A}$	$\dfrac{\cos A}{\sqrt{1-\cos^2 A}}$	$\dfrac{1}{\tan A}$	$\dfrac{1}{\sqrt{\sec^2 A - 1}}$

Worked Model: Expressing Ratios in terms of $\sec A$ (NCERT Standard)

Task: Write all other trigonometric ratios of $\angle A$ in terms of $\sec A$.

Solution: Using $\cos A = \frac{1}{\sec A}$, $\sin A = \sqrt{1-\cos^2 A} = \sqrt{1-\frac{1}{\sec^2 A}} = \frac{\sqrt{\sec^2 A - 1}}{\sec A}$, $\tan A = \sqrt{\sec^2 A - 1}$, $\cot A = \frac{1}{\tan A} = \frac{1}{\sqrt{\sec^2 A - 1}}$, and $\csc A = \frac{1}{\sin A} = \frac{\sec A}{\sqrt{\sec^2 A - 1}}$.

CHAPTER 8

WORKED MODEL

IDENTITY PROOF STRATEGIES

The 4 Cardinal Techniques & High-Scoring Board Archetypes

Module 1 | The Four Cardinal Proof Strategies

The Maths Manthan Universal Identity Solving Protocol

1. **Strategy 1 (Sines & Cosines Default):** Convert compound functions $(\tan, \cot, \sec, \csc)$ into pure sines and cosines. This resolves over 70% of board problems into elementary fraction addition.
2. **Strategy 2 (Conjugate Rationalization):** When you see denominators like $(1 \pm \sin A)$ or $(1 \pm \cos A)$ or $(\sec A \pm \tan A)$, multiply numerator and denominator by the conjugate to produce a difference of squares!
3. **Strategy 3 (Algebraic Factoring):** Look for $a^2 - b^2 = (a-b)(a+b)$ and $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$.
4. **Strategy 4 (Dual-Sided Simplification):** When direct LHS $\to$ RHS path is algebraically opaque, reduce LHS to an intermediate simplest form M, then reduce RHS independently to the exact same form M.

Module 2 | Worked Board Archetype 1: Radical Conjugate Identity

Worked Classic Model: Square Root Conjugate Rationalization

Problem: Prove that: $\sqrt{\frac{\mathbf{1+\sin A}}{\mathbf{1-\sin A}}} = \sec \mathbf{A} + \tan \mathbf{A}$.

Proof:

Step 1: Multiply numerator and denominator under radical by conjugate $(1 + \sin A)$.

$$\text{LHS} = \sqrt{\frac{1+\sin A}{1-\sin A} \times \frac{1+\sin A}{1+\sin A}} = \sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}}$$

Step 2: Apply Pythagorean identity $1 - \sin^2 A = \cos^2 A$.

$$\text{LHS} = \sqrt{\frac{(1+\sin A)^2}{\cos^2 A}} = \frac{1+\sin A}{\cos A}$$

Step 3: Split the fraction into independent terms.

$$\text{LHS} = \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \tan A = \text{RHS} \qquad \textbf{[Hence Proved]}$$

Module 3 | Worked Board Archetype 2: The $\csc A + \cot A$ Identity Transformation

Worked Classic Model: Dividing by Sine & Factoring Unity

Problem: Prove that: $\frac{\cos \mathbf{A} - \sin \mathbf{A} + \mathbf{1}}{\cos \mathbf{A} + \sin \mathbf{A} - \mathbf{1}} = \csc \mathbf{A} + \cot \mathbf{A}$ using the identity $\csc^2 A = 1 + \cot^2 A$.

Proof:

Step 1: Divide every term in numerator and denominator by $\sin A$.

$$\text{LHS} = \frac{\frac{\cos A}{\sin A} - \frac{\sin A}{\sin A} + \frac{1}{\sin A}}{\frac{\cos A}{\sin A} + \frac{\sin A}{\sin A} - \frac{1}{\sin A}} = \frac{\cot A - 1 + \csc A}{\cot A + 1 - \csc A} = \frac{(\cot A + \csc A) - 1}{(\cot A - \csc A) + 1}$$

Step 2: Substitute $1 = \csc^2 A - \cot^2 A$ into the numerator.

$$\text{LHS} = \frac{(\csc A + \cot A) - (\csc^2 A - \cot^2 A)}{\cot A - \csc A + 1} = \frac{(\csc A + \cot A) - [(\csc A - \cot A)(\csc A + \cot A)]}{\cot A - \csc A + 1}$$

Step 3: Factor out the common binomial $(\csc A + \cot A)$.

$$\text{LHS} = \frac{(\csc A + \cot A)[1 - (\csc A - \cot A)]}{1 - \csc A + \cot A} = \frac{(\csc A + \cot A)(1 - \csc A + \cot A)}{1 - \csc A + \cot A} = \csc \mathbf{A} + \cot \mathbf{A} = \text{RHS}$$

CHAPTER 8

WORKED MODEL

ADVANCED IDENTITIES

Complex Algebraic Manipulations & 5-Marker Board Models

Module 1 | Worked Board Archetype 3: The Tangent-Cotangent Cubic Identity

Worked 5-Marker Masterpiece: The Cubic Factorization Route

Problem: Prove that: $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = \mathbf{1} + \sec\theta\csc\theta$.
Proof:
Step 1: Convert all functions to pure sines and cosines.

$$\text{LHS} = \frac{\frac{\sin\theta}{\cos\theta}}{1-\frac{\cos\theta}{\sin\theta}} + \frac{\frac{\cos\theta}{\sin\theta}}{1-\frac{\sin\theta}{\cos\theta}} = \frac{\frac{\sin\theta}{\cos\theta}}{\frac{\sin\theta-\cos\theta}{\sin\theta}} + \frac{\frac{\cos\theta}{\sin\theta}}{\frac{\cos\theta-\sin\theta}{\cos\theta}}$$

$$\text{LHS} = \frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} + \frac{\cos^2\theta}{\sin\theta(\cos\theta-\sin\theta)}$$

Step 2: Align denominators by factoring a minus sign from $(\cos\theta - \sin\theta)$**.**

$$\text{LHS} = \frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \frac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)}$$

Step 3: Combine over the common denominator $\sin\theta\cos\theta(\sin\theta-\cos\theta)$**.**

$$\text{LHS} = \frac{\sin^3\theta - \cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$$

Step 4: Factor numerator using difference of cubes: $a^3 - b^3 = (a-b)(a^2+ab+b^2)$**.**

$$\text{LHS} = \frac{(\sin\theta-\cos\theta)(\sin^2\theta+\sin\theta\cos\theta+\cos^2\theta)}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$$

Step 5: Cancel common factor and substitute $\sin^2\theta + \cos^2\theta = 1$**.**

$$\text{LHS} = \frac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} + \frac{\sin\theta\cos\theta}{\sin\theta\cos\theta} = \mathbf{1} + \sec\theta\csc\theta = \text{RHS} \qquad [\textbf{Hence Proved}]$$

Module 2 | Worked Board Archetype 4: Expanding Squared Binomials

Worked 4-Marker Board Model: Perfect Square Expansion

Problem: Prove that: $(\sin\mathbf{A} + \csc\mathbf{A})^{\mathbf{2}} + (\cos\mathbf{A} + \sec\mathbf{A})^{\mathbf{2}} = \mathbf{7} + \tan^{\mathbf{2}}\mathbf{A} + \cot^{\mathbf{2}}\mathbf{A}$.
Proof:
Step 1: Expand both binomial squares using $(a+b)^2 = a^2 + 2ab + b^2$**.**

$$\text{LHS} = (\sin^2 A + 2\sin A\csc A + \csc^2 A) + (\cos^2 A + 2\cos A\sec A + \sec^2 A)$$

Step 2: Apply reciprocal relations $\sin A\csc A = 1$ **and** $\cos A\sec A = 1$**.**

$$\text{LHS} = \sin^2 A + 2(1) + \csc^2 A + \cos^2 A + 2(1) + \sec^2 A$$

Step 3: Regroup $(\sin^2 A + \cos^2 A = 1)$ **and collect integer constants.**

$$\text{LHS} = (\sin^2 A + \cos^2 A) + 2 + 2 + \csc^2 A + \sec^2 A = 1 + 4 + \csc^2 A + \sec^2 A = 5 + \csc^2 A + \sec^2 A$$

Step 4: Substitute identities $\csc^2 A = 1 + \cot^2 A$ **and** $\sec^2 A = 1 + \tan^2 A$**.**

$$\text{LHS} = 5 + (1 + \cot^2 A) + (1 + \tan^2 A) = \mathbf{7} + \tan^{\mathbf{2}}\mathbf{A} + \cot^{\mathbf{2}}\mathbf{A} = \text{RHS} \qquad [\textbf{Hence Proved}]$$

CHAPTER 8

REFERENCE MATRIX

QUICK-REVISION MATRIX

Core Laws, Formula Matrix & Curriculum Clarifications

Module 1 | Comprehensive Trigonometry Formula Reference Matrix

Category	Primary Relation	Alternative Forms	Exam Application & Memory Hook
Reciprocals	$\sin\theta = 1/\csc\theta$ $\cos\theta = 1/\sec\theta$ $\tan\theta = 1/\cot\theta$	$\sin\theta \cdot \csc\theta = 1$ $\cos\theta \cdot \sec\theta = 1$ $\tan\theta \cdot \cot\theta = 1$	Product of reciprocal pairs always equals 1. Use when secant/cosecant appears in denominator. Always invert to switch between slope and co-slope.
Quotients	$\tan\theta = \sin\theta/\cos\theta$	$\cot\theta = \cos\theta/\sin\theta$	Universal Strategy: Express non-sine/cos in quotients.
Identity 1	$\sin^2\theta + \cos^2\theta = 1$	$\sin^2\theta = 1 - \cos^2\theta$ $\cos^2\theta = 1 - \sin^2\theta$	Replaces sum of squares with unity; factor $(1-\cos)(1+\cos)$.
Identity 2	$1 + \tan^2\theta = \sec^2\theta$	$\sec^2\theta - \tan^2\theta = 1$ $\tan^2\theta = \sec^2\theta - 1$	Conjugate pair: $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$.
Identity 3	$1 + \cot^2\theta = \csc^2\theta$	$\csc^2\theta - \cot^2\theta = 1$ $\cot^2\theta = \csc^2\theta - 1$	Conjugate pair: $(\csc\theta - \cot\theta)(\csc\theta + \cot\theta) = 1$.
Special Values	$\sin 30^\circ = \cos 60^\circ = 1/2$ $\sin 60^\circ = \cos 30^\circ = \sqrt{3}/2$	$\tan 45^\circ = \cot 45^\circ = 1$ $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$	Symmetry points: complementary values are equal.

Module 2 | Geometric Symmetry & Value Invariance

Scale Invariance & Triangle Symmetries

- **Similarity Invariance:** If triangle sides are doubled or halved, all 6 trigonometric ratios remain identical.
- **Complementary Acute Angle Symmetry:** In any right triangle ABC with $\angle B = 90^\circ$, acute angles are complementary: $A + C = 90^\circ$. This explains why side BC is Opposite to A but Adjacent to C:

$$\sin A = \frac{BC}{AC} = \cos C, \qquad \tan A = \frac{BC}{AB} = \cot C, \qquad \sec A = \frac{AC}{AB} = \csc C$$

Module 3 | Curriculum Clarification Alert on Deleted Topics (CBSE 2026–27)

Deleted Topics Alert: Do NOT Waste Time Memorizing Obsolete Rules

- **Deleted Topic 8.4 (Complementary Angle Ratios):** Formulas such as $\sin(90^\circ - A) = \cos A$, $\tan(90^\circ - A) = \cot A$, and $\sec(90^\circ - A) = \csc A$ are **completely deleted** from the rationalized NCERT textbook and CBSE examinations.
- **What This Means for Students:** No board question will ask you to evaluate expressions like $\frac{\sin 18^\circ}{\cos 72^\circ}$ using complementary formulas. Focus 100% of your study effort on **Algebraic Pythagorean Identities** and **Standard Angles** $(0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ)$.

CHAPTER 8

EXAM STRATEGY

BOARD STRATEGY & AUDIT

Critical Examiner Traps, Sanity Hacks & Submission Checklist

Module 1 | The Five Lethal CBSE Board Exam Traps

Examiner Trap Alerts (Where Students Lose Marks)

1. **The Orientation Swap Trap:** Finding $\cos C$ using side BC as opposite. When reference angle shifts from A to C, Opposite and Adjacent swap immediately. Always re-orient the figure mentally!
2. **The Missing k-Factor Penalty:** Stating $\tan\theta = 4/3 \implies p = 4, b = 3$. Ratios are dimensionless proportions. Failing to write $p = 4k, b = 3k$ $(k > 0)$ causes deduction in step marks.
3. **The Squaring Notation Trap:** Writing $\sin\theta^2$ instead of $\sin^2\theta$. Remember: $(\sin\theta)^2 = \sin^2\theta \neq \sin(\theta^2)$.
4. **Illegal Angle Cancellation:** Cancelling angles as algebraic factors, e.g. $\frac{\sin 60^\circ}{\sin 30^\circ} = \frac{60}{30} = 2$. Trigonometric symbols are functions, NOT multiplying terms! $\frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$.
5. **The Root-Splitting Fallacy:** Assuming $\sqrt{\sec^2 A - 1} = \sec A - 1$ or $\sqrt{1 - \sin^2 A} = 1 - \sin A$. Square roots NEVER distribute over addition or subtraction!

Module 2 | Rapid 15-Second Numerical Verification Hacks

Instant Mental Sanity Checks

- **The 45° or 30° Numerical Sanity Test:** If stuck on whether an identity is correct in an MCQ or rough check, substitute $\theta = 45^\circ$ (or 30° if $\tan$ / $\cot$ is undefined) into both LHS and RHS. They MUST yield identical values!
- **Bound Sanity Check:** Always verify that your evaluated sine and cosine values satisfy $0 \leq |\sin\theta|, |\cos\theta| \leq 1$. If you get $\sin A = 13/12$, you have inverted opposite and hypotenuse!
- **Reciprocal Sanity Check:** At 45°, $\tan 45^\circ = \cot 45^\circ = 1$. At 30°, $\tan 30^\circ < 1 < \cot 30^\circ$.

Module 3 | Maths Manthan 5-Point Pre-Submission Audit Checklist

Maths Manthan 5-Point Pre-Submission Audit Checklist

[ ] **1. Angle Orientation Verified:** Did I identify Opposite and Adjacent correctly for the specific angle demanded ($\angle A$ vs $\angle C$)?

[ ] **2. Proportionality Factor k Declared:** Did I explicitly define $k > 0$ when constructing side lengths from given ratios?

[ ] **3. Identity Parentheses Quoted:** Are all identity substitutions accompanied by reasons (e.g. "[$\because \sec^2\theta - \tan^2\theta = 1$]")?

[ ] **4. Fractions Rationalized:** Are all final answers cleared of radicals in denominators (e.g. $1/\sqrt{3} \to \sqrt{3}/3$)?

[ ] **5. LHS to RHS Closure:** Does the identity proof conclude with a clear LHS = RHS declaration and "Hence Proved"?

Maths Manthan Tuition Classes

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End of Chapter 8 Handbook – Designed for Academic Excellence in CBSE Board Examinations.